Hotel Housekeeping Management System - Maintenance and Support Document

1. System Administrator Contact Information

Item	Details
Primary Contact	Qingchao Li / Worarat Suwanwattana
Email	270758686@yoobeestudent.ac.nz /270718902@yoobeestudent.ac.nz
Support Hours	Monday-Friday, 9:00 AM - 5:00 PM
Emergency Contact	Same email (monitored during off-hours for critical issues)

2. System Monitoring Requirements

2.1 Application Health Monitoring

Item	Details
Public URL	`http://32.192.68.2/`
Nginx (Reverse Proxy)	Port 80
Backend API Server	`http://127.0.0.1:8000` (FastAPI)
Backend Internal	`http://localhost:8000` (accessed via Nginx)
Monitoring Frequency	Every 5 minutes
Swagger Docs	`http://32.192.68.2/docs`

Key Metrics to Track:

Metric	Target	Alert Threshold
API Response Time	< 500ms	> 1000ms
Error Rate	< 1%	> 5%
Frontend Build Time	< 30s	> 60s
Database Query Time	< 200ms	> 1000ms
System Uptime	> 99%	< 95%
Disk Usage	< 80%	> 90%
Memory Usage	< 80%	> 90%

2.4 System Architecture

```
                   Internet
                      │
                      ▼
               ┌──────────────┐
               │  Nginx :80   │
               │  (Proxy)     │
               └──────┬───────┘
                      │
        ┌─────────────┼─────────────┐
        │             │             │
        ▼             ▼             ▼
   ┌─────────┐  ┌────────────┐  ┌─────────┐
   │Frontend │  │ Backend API│  │ MySQL   │
   │(Static) │  │ :8000      │  │ :3306   │
   │Files    │  │ (FastAPI)  │  │         │
   └─────────┘  └────────────┘  └─────────┘
```

Health Check Commands:

```
# Overall health check
curl -I http://32.192.68.2/
curl -I http://32.192.68.2/docs
curl -s http://32.192.68.2/api/portal/services

# Backend direct check
curl -I http://127.0.0.1:8000/
curl http://127.0.0.1:8000/openapi.json
```

2.2 Database Monitoring

Item	Details
Database Name	housekeeping_new
Database Type	MySQL
Connection	Via SQLAlchemy ORM
Connection Pool	Monitor usage, alerts at 80% capacity

Monitoring Checklist:

- Connection Pool: Monitor usage and set alerts at 80% capacity
- Query Performance: Track slow queries (> 2 seconds)
- Storage Usage: Monitor disk space with alerts at 80% usage
- Backup Status: Verify daily backups complete successfully

2.3 Performance Metrics Dashboard

Metric	Target	Alert Threshold
API Response Time	< 500ms	> 1000ms
Error Rate	< 1%	> 5%
Database Connections	< 80% pool	> 90% pool
Frontend Load Time	< 3s	> 5s
CPU Usage	< 70%	> 85%
Memory Usage	< 75%	> 90%

3. Regular Maintenance Schedule

3.1 Daily Maintenance Tasks

1. System Log Review (via SSH)

```
ssh root@32.192.68.2

# Check backend logs
tail -f /var/log/housekeeping.log
grep -i error /var/log/housekeeping.log

# Check Nginx access logs
tail -f /var/log/nginx/access.log
tail -f /var/log/nginx/error.log
```

- Check application error logs
- Review authentication failure attempts
- Monitor suspicious activity patterns

2. Backup Verification

```
ssh root@32.192.68.2

# Check backup directory
ls -la /backups/database/
ls -lh /backups/database/*.sql.gz | head -7

# Verify latest backup
zcat /backups/database/housekeeping_$(date +%Y%m%d)*.sql.gz 2>/dev/null | head -5
```

- Confirm database backup completion
- Verify backup file integrity
- Check backup storage availability

3. Performance Check

```bash
ssh root@32.192.68.2

# Test API response
curl -o /dev/null -s -w "Time: %{time_total}s\n" http://127.0.0.1:8000/api/portal/servi
curl -o /dev/null -s -w "Time: %{time_total}s\n" http://32.192.68.2/api/portal/services

# Check system resources
df -h
free -h
```

- Review API response times
- Check database connection health
- Monitor system resource usage

3.2 Weekly Maintenance Tasks

1. Security Audit

```bash
ssh root@32.192.68.2

# Review Nginx access logs for suspicious activity
grep -E "500|401|403" /var/log/nginx/access.log | tail -20
grep -i "sqlmap\|python\|curl" /var/log/nginx/access.log

# Check for unauthorized SSH attempts
grep "Failed password" /var/log/auth.log | tail -10

# Review recent git commits
cd /path/to/project && git log --oneline -7
```

- Review access logs
- Check for unauthorized access attempts
- Update security patches if available

2. Database Maintenance

```
ssh root@32.192.68.2

mysql -h localhost -u root -p -e "
SELECT table_name,
       ROUND(data_length / 1024 / 1024, 2) 'Data (MB)',
       ROUND(index_length / 1024 / 1024, 2) 'Index (MB)'
FROM information_schema.tables
WHERE table_schema = 'housekeeping_new'
ORDER BY data_length DESC;
"
```

- Analyze query performance
- Check and optimize indexes
- Review table sizes and growth

3. User Management

```
ssh root@32.192.68.2

mysql -h localhost -u root -p -e "
SELECT id, username, email, create_time, is_deleted
FROM housekeeping_new.user
WHERE is_deleted = 0
ORDER BY create_time DESC LIMIT 10;
"
```

- Review new user registrations
- Check for suspicious accounts
- Audit admin user activities

3.3 Monthly Maintenance Tasks

1. System Updates

```
ssh root@32.192.68.2

# Navigate to project
cd /path/to/project

# Pull latest code
git pull origin main

# Backend updates
cd backend/src
pip install -r requirements.txt --upgrade

# Frontend production build
cd ../frontend
npm install
npm run build

# Copy new build to Nginx
sudo cp -r dist/* /var/www/housekeeping/dist/

# Restart backend if needed
pkill -f "uvicorn app:app"
cd /path/to/backend/src && nohup python -m uvicorn app:app --host 0.0.0.0 --port 8000 >
```

- Update Python dependencies
- Apply security patches
- Review and update configuration

2. Performance Optimization

```
ssh root@32.192.68.2

# Check disk space
df -h

# Check database size growth
mysql -h localhost -u root -p -e "
SELECT table_schema 'Database',
       ROUND(SUM(data_length + index_length) / 1024 / 1024 / 1024, 2) 'Size (GB)'
FROM information_schema.tables
WHERE table_schema = 'housekeeping_new'
GROUP BY table_schema;
"

# Optimize database tables
mysql -h localhost -u root -p -e "OPTIMIZE TABLE housekeeping_new.service_order;"
```

- Analyze system bottlenecks
- Optimize database queries
- Review caching strategies

3. Capacity Planning

- Review usage trends
- Plan for scaling needs
- Update backup and recovery procedures

4. Troubleshooting Guide

4.1 Common Issues and Solutions

Issue 1: Site Not Accessible (Port 80)

Symptoms:

- Error: "This site can't be reached"
- Nginx 502 Bad Gateway errors

Solutions:

```bash
# SSH to server
ssh root@32.192.68.2

# Check Nginx status
sudo systemctl status nginx
sudo nginx -t

# Check if port 80 is listening
lsof -i :80

# Restart Nginx
sudo systemctl restart nginx
```

Issue 2: Backend API Not Working

Symptoms:

- 502 Bad Gateway from Nginx
- API requests failing
- Backend errors in logs

Solutions:

```bash
# SSH to server
ssh root@32.192.68.2

# Check backend process
ps aux | grep uvicorn
lsof -i :8000

# Check backend logs
tail -100 /var/log/housekeeping.log

# Restart backend
pkill -f "uvicorn app:app"
cd /path/to/backend/src && nohup python -m uvicorn app:app --host 0.0.0.0 --port 8000 >

# Test backend directly
curl http://127.0.0.1:8000/docs
```

Issue 3: Database Connection Problems

Symptoms:

- "Cannot connect to database" errors
- Slow query responses
- Connection timeout errors

Solutions:

```
# SSH to server
ssh root@32.192.68.2

# Verify database service is running
sudo systemctl status mysql

# Check if MySQL is listening
lsof -i :3306

# Test database connection
mysql -h localhost -u root -p -e "SELECT 1;"

# Check database configuration
cat /path/to/backend/.env | grep DB_

# Verify database exists
mysql -h localhost -u root -p -e "SHOW DATABASES LIKE 'housekeeping_new';"

# Check database size
mysql -h localhost -u root -p -e "SELECT table_schema 'Database', ROUND(SUM(data_length
```

Issue 4: Authentication Failures

Symptoms:

- "Invalid credentials" errors
- JWT token expiration issues
- Role permission errors

Solutions:

```
# SSH to server
ssh root@32.192.68.2

# Check user exists in database
mysql -h localhost -u root -p -e "SELECT id, username, email, status FROM housekeeping_

# Check JWT configuration
cat /path/to/backend/.env | grep -i jwt
cat /path/to/backend/src/core/security.py | grep -i secret

# Test login directly on backend
curl -X POST http://127.0.0.1:8000/api/auth/login/json \
  -H "Content-Type: application/json" \
  -d '{"username":"guest","password":"admin123"}'
```

Default Test Accounts:

Role	Username	Password
Admin	admin	admin123
Guest	guest	admin123
Cleaner	cleaner1	admin123

4.2 Error Code Reference

Error Code	Description	Recommended Action
500	Internal Server Error	Check application logs, restart service
401	Unauthorized	Verify credentials, check token expiration
403	Forbidden	Check user role permissions
404	Not Found	Verify endpoint URL, check resource existence
422	Validation Error	Review request data format
502	Bad Gateway	Check backend service status
503	Service Unavailable	Restart backend service

4.3 Frontend-Specific Issues (Production)

Issue	Solution
Frontend not loading	Check Nginx config and static files location
Vue Router 404 errors	Update Nginx try_files directive
API calls returning 404	Check Nginx proxy_pass configuration
CORS errors	Verify backend CORS settings
Static files not found	Verify /var/www/housekeeping/dist/ exists

Frontend Troubleshooting:

```
ssh root@32.192.68.2

# Check Nginx static file config
cat /etc/nginx/conf.d/housekeeping.conf

# Verify static files exist
ls -la /var/www/housekeeping/dist/
ls -la /var/www/housekeeping/dist/*.html

# Check Nginx error logs
tail -20 /var/log/nginx/error.log

# Reload Nginx
sudo nginx -s reload
```

5. Backup and Recovery Procedures

5.1 Database Backup

```bash
#!/bin/bash
# Daily automated backup script for housekeeping_new database
# Server: 32.192.68.2

BACKUP_DIR="/backups/database"
DATE=$(date +%Y%m%d_%H%M%S)
BACKUP_FILE="$BACKUP_DIR/housekeeping_$DATE.sql"

# Create backup (database on same server)
mysqldump --single-transaction --routines --triggers \
  -h localhost -u root -p'your_password' housekeeping_new > $BACKUP_FILE

# Compress backup
gzip $BACKUP_FILE

# Keep only last 7 days of backups
find $BACKUP_DIR -name "*.sql.gz" -mtime +7 -delete

# Remote backup copy (optional - to another server)
# rsync -av $BACKUP_DIR backup_server:/backups/

echo "Backup completed: $BACKUP_FILE.gz"
```

5.2 Application Backup

1. Source Code Backup:

```bash
# Create backup of entire project
tar -czf backup_$(date +%Y%m%d).tar.gz \
  --exclude='node_modules' \
  --exclude='__pycache__' \
  --exclude='*.pyc' \
  assessment2/
```

2. Configuration Backup:

```
# Backup environment files
cp backend/.env backend/.env.backup
cp frontend/.env frontend/.env.backup
```

3. Uploaded Files Backup:

```
# Backup any uploaded files
tar -czf uploads_backup_$(date +%Y%m%d).tar.gz uploads/
```

5.3 Recovery Procedures

Database Recovery:

```
# Restore from backup
gunzip -c backup_file.sql.gz | mysql -h localhost -u root -p'your_password' housekeepin
```

Application Recovery:

1. Restore from git repository
2. Reinstall dependencies
3. Restore configuration files
4. Start application services
5. Run health checks

6. Security Maintenance

6.1 Regular Security Tasks

1. Patch Management:

- Weekly review of security updates
- Monthly application of critical patches
- Quarterly comprehensive security review

2. Access Control:

```
# Review recent logins
mysql -u root -p -e "SELECT * FROM housekeeping_new.user WHERE is_deleted = 0 ORDER BY
```

- Monthly review of user permissions
- Quarterly audit of admin access
- Immediate revocation of inactive accounts

3. Data Protection:

- Encrypt sensitive data at rest
- Secure data transmission (HTTPS in production)
- Regular security scanning

6.2 Incident Response

Security Incident Classification:

Level	Classification	Response
Critical	Unauthorized data access	Immediate action
High	System breach attempts	Within 1 hour
Medium	Suspicious activity	Within 4 hours
Low	Minor security warnings	Within 24 hours

Response Procedures:

1. Immediate isolation of affected systems
2. Preservation of logs and evidence
3. Notification to stakeholders
4. Root cause analysis and remediation

7. Performance Optimization

7.1 Monitoring Points

API Endpoint Performance:

Endpoint	Description	Performance Target
`/api/auth/login/json`	User authentication	< 500ms
`/api/portal/services`	Service catalog	< 300ms
`/api/portal/order`	Create order	< 500ms
`/api/portal/requirements`	Browse requirements	< 300ms

Database Performance:

- Query execution time
- Index utilization
- Connection pool efficiency

7.2 Optimization Actions

Database Optimization:

```sql
-- Check slow queries
SHOW VARIABLES LIKE 'slow_query_log';
SHOW VARIABLES LIKE 'long_query_time';

-- Analyze tables
ANALYZE TABLE service_order;
ANALYZE TABLE user;
ANALYZE TABLE customer_requirement;
```

Application Optimization:

```bash
# Frontend build optimization
npm run build  # Production build

# Check for unused dependencies
npm audit
```

8. Support Request Process

8.1 Contact Information

Item	Details
Primary Support	270758686@yoobeestudent.ac.nz
Response Time	Within 4 business hours
Emergency Support	Same email (24/7 monitoring for critical issues)

8.2 Support Request Format

When submitting a support request, please include:

1. **Issue Description**: Detailed explanation of the problem
2. **Error Messages**: Copy of any error messages
3. **Steps to Reproduce**: Exact steps to recreate the issue
4. **System Information**:
    - Python version
    - Node.js version
    - Database type (MySQL)
    - OS
5. **Timestamp**: When the issue occurred

8.3 Support Levels

Level	Response Time	Resolution Time	Examples
Critical	1 hour	4 hours	System down, data loss
High	4 hours	24 hours	Major functionality broken
Medium	8 hours	48 hours	Minor bugs, performance issues
Low	24 hours	5 days	Feature requests, documentation

9. System Recovery Procedures

9.1 Disaster Recovery Plan

1. Immediate Actions (First 30 minutes):

- Assess impact and scope
- Activate backup team
- Begin recovery procedures

2. Recovery Phase (30 mins - 4 hours):

```
# Check system status on server 32.192.68.2
ssh root@32.192.68.2 "lsof -i :80"
ssh root@32.192.68.2 "lsof -i :8000"
ssh root@32.192.68.2 "ps aux | grep -E '(uvicorn|nginx)'"

# Restart Nginx
ssh root@32.192.68.2 "sudo systemctl restart nginx"

# Restart Backend
ssh root@32.192.68.2 "pkill -f 'uvicorn app:app'; cd /path/to/backend/src && nohup pyth

# Verify services
curl -I http://32.192.68.2/
curl -I http://32.192.68.2/docs
```

- Restore from latest backup
- Verify data integrity
- Test critical functionality

3. Restoration Phase (4-24 hours):

- Full system restoration
- Data synchronization
- User notification

9.2 Recovery Testing

- Monthly backup restoration test
- Quarterly disaster recovery drill

- Annual comprehensive recovery test

10. System Configuration Reference

10.1 Server Information

Item	Details
Server IP	32.192.68.2
Server Type	Cloud Server (all-in-one deployment)
Public Access	http://32.192.68.2
SSH Access	ssh root@32.192.68.2

10.2 Service Ports

Service	Port	Description	Access
Nginx	80	Reverse proxy, serves frontend	Public
Backend API	8000	FastAPI application	Internal only
MySQL	3306	Database server	Localhost only

10.3 Nginx Configuration

```
# /etc/nginx/conf.d/housekeeping.conf
server {
    listen 80;
    server_name 32.192.68.2;

    # Backend API proxy
    location /api/ {
        proxy_pass http://127.0.0.1:8000/api/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }

    # Frontend static files
    location / {
        root /var/www/housekeeping/dist;
        try_files $uri $uri/ /index.html;
    }
}
```

10.4 Backend Environment Variables

Location: `/path/to/backend/.env`

```
DB_USER=root
DB_PASSWORD=your_password
DB_HOST=localhost
DB_PORT=3306
DB_NAME=housekeeping_new
```

10.5 Service Management Commands

```
# Nginx
sudo systemctl status nginx
sudo systemctl restart nginx
sudo nginx -t  # Test configuration

# Backend (using systemd or supervisor)
sudo systemctl status housekeeping
sudo systemctl restart housekeeping

# Or manual process management
pkill -f "uvicorn app:app"
cd /path/to/backend/src && nohup python -m uvicorn app:app --host 0.0.0.0 --port 8000 >

# MySQL
sudo systemctl status mysql
sudo systemctl restart mysql
```

10.3 Default Credentials

Role	Username	Password
Admin	admin	admin123
Guest	guest	admin123
Cleaner	cleaner1	admin123

10.6 Key API Endpoints

Base URL: `http://32.192.68.2/api/`

Endpoint	Method	Description
`/api/auth/login/json`	POST	User login
`/api/auth/register`	POST	User registration
`/api/auth/verify-2fa`	POST	Two-factor verification
`/api/portal/services`	GET	List service types

Endpoint	Method	Description
`/api/portal/services/{id}`	GET	Service type detail
`/api/portal/order`	POST	Create order
`/api/portal/orders/{phone}`	GET	Get orders by phone
`/api/portal/requirements`	GET	Browse requirements
`/api/portal/cleaners`	GET	List cleaners
`/api/portal/cleaners/{id}`	GET	Cleaner detail
`/api/portal/apply`	POST	Apply for requirement
`/api/complaint`	POST	Submit complaint
`/api/wallet/balance`	GET	Get wallet balance

10.7 Deployment Checklist

When deploying updates to production:

```
ssh root@32.192.68.2

# 1. Backup database before update
mysqldump --single-transaction -h localhost -u root -p housekeeping_new > /backups/data

# 2. Pull latest code
cd /path/to/project && git pull

# 3. Backend update
cd backend/src
pip install -r requirements.txt
pkill -f "uvicorn app:app"
nohup python -m uvicorn app:app --host 0.0.0.0 --port 8000 >> /var/log/housekeeping.log

# 4. Frontend build
cd ../../frontend
npm install
npm run build
sudo cp -r dist/* /var/www/housekeeping/dist/

# 5. Verify
curl http://32.192.68.2/docs
curl http://32.192.68.2/api/portal/services
```

11. End of Support Information

11.1 Support Timeline

Item	Duration
Active Support	6 months from submission date
Security Updates	Provided for active support period
Bug Fixes	Addressed based on severity during active support

11.2 Transition Plan

1. Documentation Handover:

- Complete system documentation
- Operational procedures
- Troubleshooting guides

2. Knowledge Transfer:

- System architecture overview
- Maintenance procedures
- Common issues and solutions

3. Support Transition:

- Gradual reduction of support
- Final knowledge transfer session
- Handover completion

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Deployment: 32.192.68.2 (All-in-one: Nginx + FastAPI + MySQL)